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Experimental and simulation study on wettability of high temperature liquid sodium on metal plate surface

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Highlights

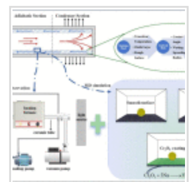
Outline

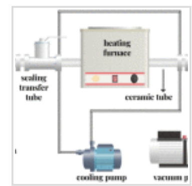
- Combined experiments and MD simulations are performed to link macroscopic contact angles with atomic-scale interfacial dynamics.
- Revealed critical temperature thresholds governing sodium-stainless steel interactions through oxide reduction.
- Demonstrated oxide coatings suppress precursor film formation, reducing wetting radii by 10.44–15.56% versus bare surfaces.
- Established oxide coatings maintain higher contact angles while roughness enhances Wenzel-mode wetting.

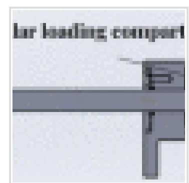
Abstract

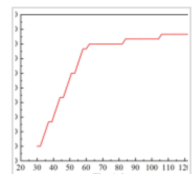
Interfacial wetting behavior between liquid sodium and stainless steel is one of the fundamental concerns in high-temperature liquid metal systems, with critical implications for heat transfer efficiency, corrosion resistance, and material compatibility in nuclear energy and thermal storage applications. This study systematically investigates the interfacial wetting mechanisms through combined approaches of experiments and simulations. The wetting characteristics of liquid sodium on both bare and Cr_2O_3-coated stainless steel surfaces are examined in experiments. Furthermore, the MD simulations are carried out to explore wetting kinetics on smooth stainless steel, Cr_2O_3-coated surfaces, and rectangular grooved surfaces. Two scenarios of isothermal and stepwise heating are simulated to analyze transient variations in contact angle, precursor film formation, and equivalent wetting radius. The temperature critically governs wettability: no wetting occurs between liquid sodium and stainless steel below 300°C, while the transition point is approximately 320–335°C. The Cr_2O_3 coating deteriorates interfacial wettability, yielding consistently higher contact angles compared to uncoated surfaces. Rough surface follows the Wenzel model, showing suppressed spreading but accelerated contact angle reduction at elevated temperatures. These insights clarify the synergistic effects of temperature, oxide layers and surface topography in high-temperature sodium

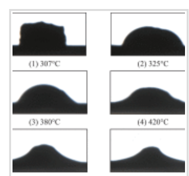
Figures (19)

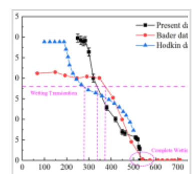












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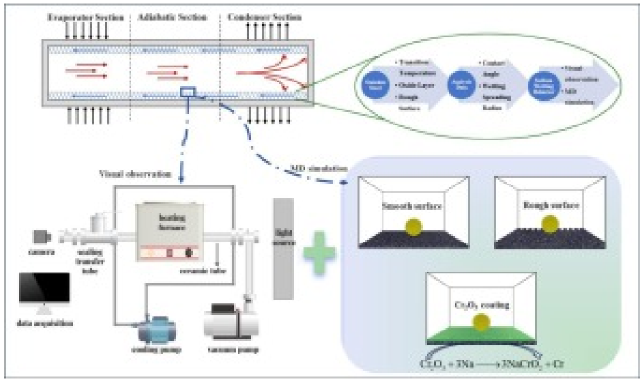
environments, providing actionable insights for optimizing material design and operational reliability.

Graphical abstract



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Keywords

Liquid sodium; Oxide layer; Wettability; Contact angle; Molecular dynamics

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1. Introduction

Interfacial wetting behavior between liquid sodium and stainless steel represents a fundamental challenge in advanced energy systems, with critical implications for heat transfer efficiency, material compatibility, and operational safety in next-generation nuclear reactors and thermal energy storage technologies. As global energy systems undergo profound transformation toward carbon neutrality, nuclear energy emerges as a uniquely reliable clean energy source – offering stable baseload power generation with minimal geographical constraints compared to intermittent renewables like solar and wind. The optimization of liquid metal interface phenomena, particularly in sodium-cooled fast reactors (SFRs) and heat pipe-cooled reactors (HPCRs), thus becomes paramount for enhancing energy conversion efficiency while ensuring long-term material durability in these mission-critical systems [[1], [2], [3]].

Space nuclear power systems, which combine nuclear energy with advanced thermoelectric conversion technologies, offer unparalleled advantages for extreme environment applications in marine and aerospace missions [4]. Among various configurations, HPCRs have emerged as the leading solution for future space nuclear applications due to their innovative heat transfer mechanism. HPCRs employ high-temperature heat pipes to efficiently extract thermal energy from the reactor core, enabling a compact system design while ensuring inherent safety and reliability. Alkali metals such as sodium, potassium, and lithium are commonly employed as working fluids in these systems due to their exceptional thermal conductivity and high boiling points [5]. Within the heat pipe, the liquid alkali metal absorbs thermal energy and vaporizes in the evaporator section. The resulting vapor subsequently flows to the condenser section, where it releases latent heat through condensation. Capillary action generated by the porous wick structure then drives the condensate back to the evaporator, completing a self-sustaining, passive heat transfer cycle. Operating at elevated temperatures, these heat pipes provide outstanding thermal performance through their high heat transfer efficiency, simple structural configuration, and remarkable temperature uniformity – critical attributes for space-based nuclear systems. This modular design ensures that each heat pipe operates as an autonomous thermal conduction unit, inherently eliminating single-point failure risks in the system [6].

Nevertheless, sodium's inherent properties as an alkali metal present notable challenges. Its high chemical reactivity with both water and oxygen necessitates stringent containment measures. More critically, under elevated operating temperatures, sodium can engage in complex interfacial phenomena with structural materials. These interactions profoundly influence the liquid metal's wetting characteristics and spreading dynamics on material surfaces, potentially impacting overall system performance and reliability [[7], [8], [9]]. Borgstedt et al. [10] studied the corrosion of reactor materials in high-temperature flowing sodium. Corrosion tests on fuel element materials from large-capacity sodium-cooled reactors showed that high-quality austenitic steels and nickel-based alloys exhibited relatively low sodium corrosion. Therefore, structural materials in reactors often applied stainless steels with good strength and corrosion resistance. Many studies have found that factors such as the temperature and flow rate of liquid sodium, as well as the concentration of various impurities in the sodium, also influenced the wetting and corrosion behavior of stainless steel [11,12]. Studies have shown that the dynamic spreading behavior of high-temperature liquid metals on solid surfaces often deviated significantly from classical theoretical predictions, with experimentally observed spreading rates being several orders of magnitude lower than theoretical values. The presence of oxide films on the liquid metal surface, substrate roughness, and weak interfacial reactions were identified as key factors causing this deviation [13]. Based on the above analysis, investigating the spreading kinetics of liquid sodium on stainless steel surfaces under different surface conditions during actual operation is crucial for improving heat pipe performance.

While significant progress has been made in understanding the wettability of liquid sodium on metallic surfaces, critical knowledge gaps persist regarding fundamental interfacial parameters. The contact angle, serving as a primary quantitative indicator of wettability, remains insufficiently characterized for liquid sodium-metal systems under operational conditions. The Young equation describes the intrinsic contact angle on an ideal smooth surface [14]. The Wenzel model [15] and the Cassie-Baxter model [16] introduce the effects of surface roughness and composite gas–liquid interfaces on the apparent contact angle, respectively. The measurement of contact angles primarily employs three established techniques: the sessile drop method, the flat plate method [17], and the capillary rise method [18]. These approaches are typically coupled with advanced image analysis and computational fitting algorithms to enhance measurement accuracy and reliability. Faizal et al. [19] developed the contact angle measurement system within an inert atmosphere chamber to measure the variation of the contact angle of liquid sodium on SS316LN with changing temperature. Hodkin et al. [20] evaluated the wettability of liquid sodium on various metals and alloys using a sessile drop method system. They investigated the effects of key factors such as substrate surface treatment, oxygen content in the sodium, and material composition. Their study found that the surface chromium-to-oxygen ratio has a significant impact on wettability behavior. However, Barkia et al. [21] confirmed through transmission electron microscopy that pre-exposure of 304L steel to oxygen-containing liquid sodium leads to the formation of a Na_xCrO_2 interfacial layer, which significantly enhances wettability. They also emphasized the importance of the oxide film wetting mechanism. Saito et al. [22] used the tangent method combining experimental and theoretical analyses to determine contact angles for developing the correlation between the contact angle and atomic bonding. Bader et al. [23] studied the wettability of liquid sodium on 304L stainless steel and found that the contact angle decreases with increasing temperature, eventually stabilizing after 1 to 2 h. Current experimental studies on sodium wettability primarily focused on macroscopic performance, while overlooking the microscopic mechanisms of sodium wetting and spreading behavior [24]. Furthermore, technical limitations in earlier experiments often led to alkali metal contamination, resulting in insufficient accuracy and mechanistic understanding of how stainless steel surface oxides affect liquid sodium wettability [25].

In the field of numerical simulation, some studies have further revealed the kinetic mechanisms of droplet spreading. Liang et al. [26] investigated the wettability of liquid sodium on nine transition metals using SEM, XRD and MD simulations. They found that metals with face-centered cubic or hexagonal close-packed structures generally exhibit better spreading performance than metals with body-centered cubic structures. Tong et al. [27], using droplet impact simulations, pointed out that impact velocity, viscosity, and surface tension were the primary parameters influencing the spreading factor. Additionally, they found that substrate temperature indirectly regulated the spreading distance by affecting the rate of heat transfer. Misyura et al. [28] used MD simulations to model the wettability in metal matrix composites and investigated the sensitivity of the wettability to surface temperature for a variety of materials, such as polished as well as laser texturized treatments with or without pits. Nevertheless, existing simulation studies predominantly examined individual surface types, failing to compare wettability across oxidized or textured stainless steel surfaces. More critically, no systematic investigation has addressed liquid sodium spreading dynamics under realistic heat pipe operating conditions.

This study presents a multiscale investigation of high-temperature liquid sodium wettability by integrating macroscopic visualization experiments with molecular dynamics simulations. Three representative surface types are systematically examined: smooth stainless steel, Cr_2O_3-coated stainless steel with varying thicknesses, and rectangular grooved surfaces. Through quantitative characterization of dynamic wetting parameters including contact angle evolution, precursor film formation, and equivalent wetting radius, the fundamental mechanisms governing interfacial wetting behavior across different surface morphologies are investigated.

2. Experiment design

2.1. Experimental facility

The wettability experiments of liquid sodium on metal surfaces are performed in a custom-designed high-temperature vacuum facility. As illustrated in Fig. 1, the experimental system features a horizontal alumina ceramic tube with its central section enclosed by a resistance heating furnace, capable of achieving precise temperature control. The system configuration includes: (1) a sealing transfer device connected to the tube's left flange for sample introduction, (2) opposed optical access ports with backlighting and imaging, (3) active cooling circuits for thermal management of vacuum seals, (4) a high-vacuum pumping system with molecular pumps, and (5) a computerized data acquisition system for synchronized temperature and imaging data collection.

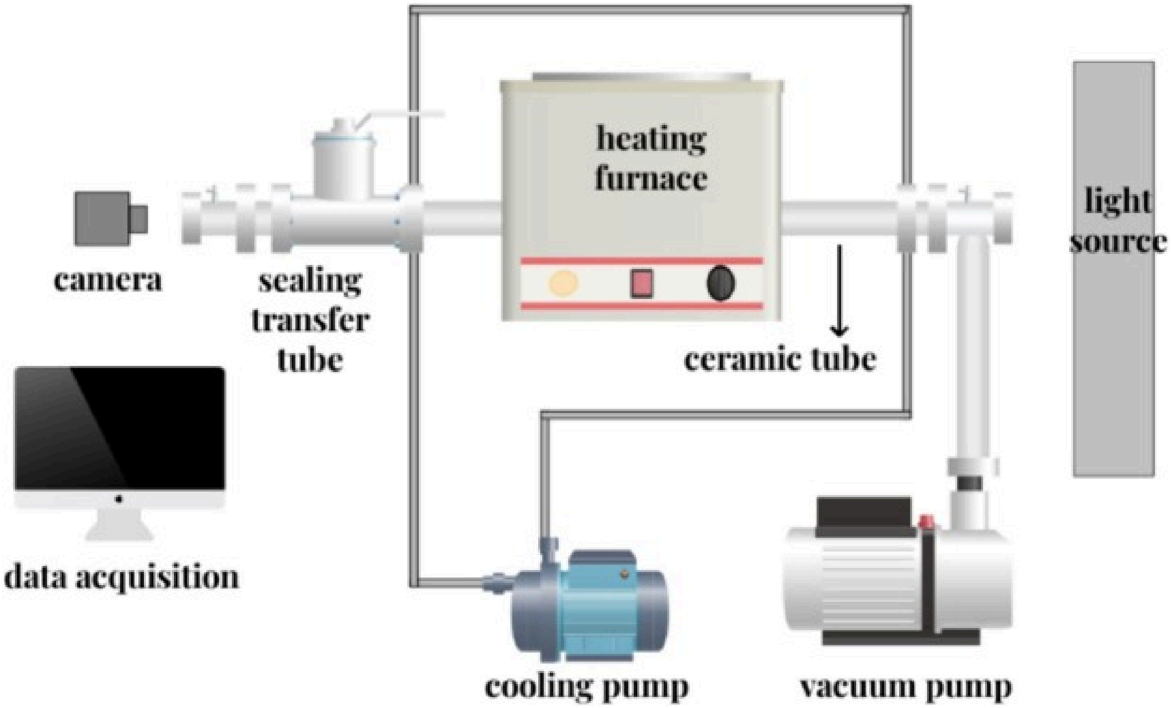

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Fig. 1. Schematic diagram of the wettability facility.

The heating system consists of a muffle furnace integrated with a horizontal alumina ceramic tube, capable of achieving temperatures up to 1600°C. The test specimens (sodium block and metal plate) inside the ceramic tube, with temperature monitoring accomplished using centrally located K-type thermocouples. To maintain system integrity during high-temperature operation, dual vacuum flanges at both ends of the ceramic tube are actively cooled by the circulating water system. The vacuum conditions are achieved using Edwards T85 molecular turbopump, which can establish an ultra-high vacuum environment down to 10^{-5} Pa within the chamber. The imaging system incorporates a CMOS digital camera coupled with a high-resolution zoom microscope and high-intensity LED illumination, with all images processed through specialized computer software capable of contact angle measurements with±0.1° accuracy through multiple advanced fitting algorithms.

Due to its highly reactive physical and chemical properties, alkali metal sodium is easily contaminated by steam and oxygen in the air at room temperature, and can also burn with oxygen and nitrogen at high temperatures. Therefore, it is necessary to strictly control the concentrations of steam, nitrogen and oxygen in the experimental environment. Then we controlled the vacuum degree inside the ceramic tube and filled it with argon gas for protection. In addition, there may also be oxidation issues during the process of transferring the sodium blocks prepared in the glove box to the desired location in the ceramic tube. Therefore, a sealing transfer device is designed that can maintain the entire process oxygen free, as shown in Fig. 2.

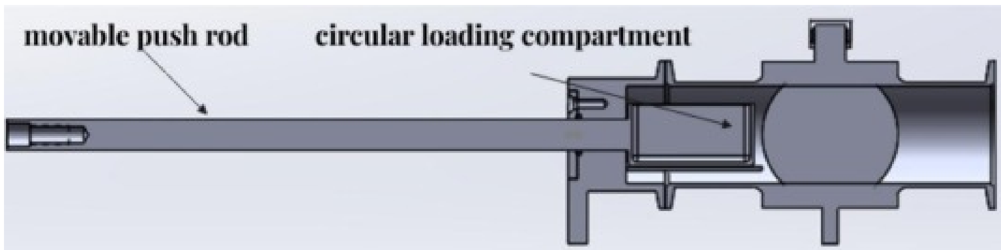

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Fig. 2. Schematic diagram of the sealing transfer device.

The main body of the device is designed as a ball valve integrated with a cylindrical loading compartment and a movable push rod. The former is used for storing prepared sodium block and the latter is characteristized with dynamic sealing. Prior to operation, the entire device is placed in the glove box with the concentrations of oxygen and vapor to be less than 0.5 ppm and 0.1 ppm respectively. The sodium block is also cut inside the glove box into the size of 2 mm cube shape, taking into account the influence of sodium evaporation and contact angle measurement standards. The prepared sodium block is loaded into the circular compartment and hermetically sealed. The device is then flangemounted to the left port of ceramic tube for sample deployment. The ball valve is opened to establish continuous inert atmosphere and the push rod advances the sodium cube to the predetermined deposition position, allowing gravitational placement onto the substrate. Afterwards, the movable rod is retracted and the ball valve is closed. At last, the objective lens chamber is reconnected and the ball valve is opened again to proceed the experiment.

2.2. Experimental procedure

Due to the experimental conditions and environmental requirements, the operation process can be divided into the following six parts in order: equipment inspection; metal plate processing and placement; loading and transfer of sodium blocks; heating setup and data collection; heating end and sample extraction; data analysis and processing.

Before the experiment, it is necessary to check the airtightness of the glove box and the wettability facility. The concentrations of steam and oxygen are automatic monitored. For the metal plate processing, stainless steel with different oxidation thickness are selected as the test metal plate, which is in flat square shape with the side length of 49 mm. The surface is cleaned using ultrasonic cleaning device with anhydrous ethanol and acetone solution sequentially. The plate is placed horizontally in the ceramic tube at the locations of heating furnace chamber. The process of sodium block transfer is conducted by sealing transfer device, as described above. After the sodium block is placed on the flat plate, the heating of the ceramic tube and the cooling of the flanges are turned on. The photos of cross section inside the visual window at different temperature levels are taken and the images are analyzed to obtain the information of contact angles.

Previous studies have demonstrated that temperature significantly influences the contact angle of liquid alkali metals. For liquid sodium on stainless steel surfaces, complete wetting is typically achieved below 600 °C [23]. To ensure experimental accuracy, all tests are conducted under a controlled vacuum of 0.06 MPa, which can minimize impurity gas contamination and maintain sodium boiling point above 600 °C to prevent significant volatilization. As illustrated in Fig. 3, the stepped heating protocol is implemented in the test, with temperature stabilization at designated intervals to obtain reliable equilibrium wetting data.

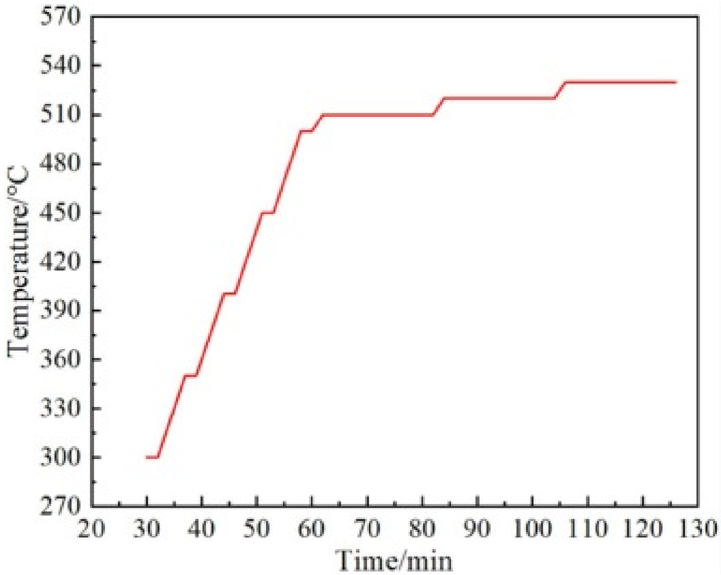

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Fig. 3. Heating program during the experiment.

3. Experimental results

3.1. Experimental results of the stainless steel plate

For the test with 304 stainless steel, the variation of images of sodium appearance on the plate surface during the heating stage is shown in Fig. 4. Initially, at temperatures around 307°C, the sodium block maintains its solid form due to surface oxidation, where a protective Na_2O crust (melting point: 1132°C) encapsulates the molten sodium core (melting point: 97.7°C), preventing any observable wetting. As temperature rises to 325°C, thermal stress induces sudden rupture of the oxide layer, releasing the liquid sodium which then assumes a characteristic hemispherical shape. With further temperature increase, the system undergoes a gradual wetting transition, evidenced by progressive reduction in contact angle below 90°. When stabilized at 530 °C for 20 min, the liquid sodium achieves complete wetting, demonstrating excellent interfacial adhesion with the stainless steel substrate. This temperature-dependent wetting transition highlights the crucial interplay between thermal energy and oxide layer dynamics in determining the interfacial behavior of liquid sodium on metal surfaces.

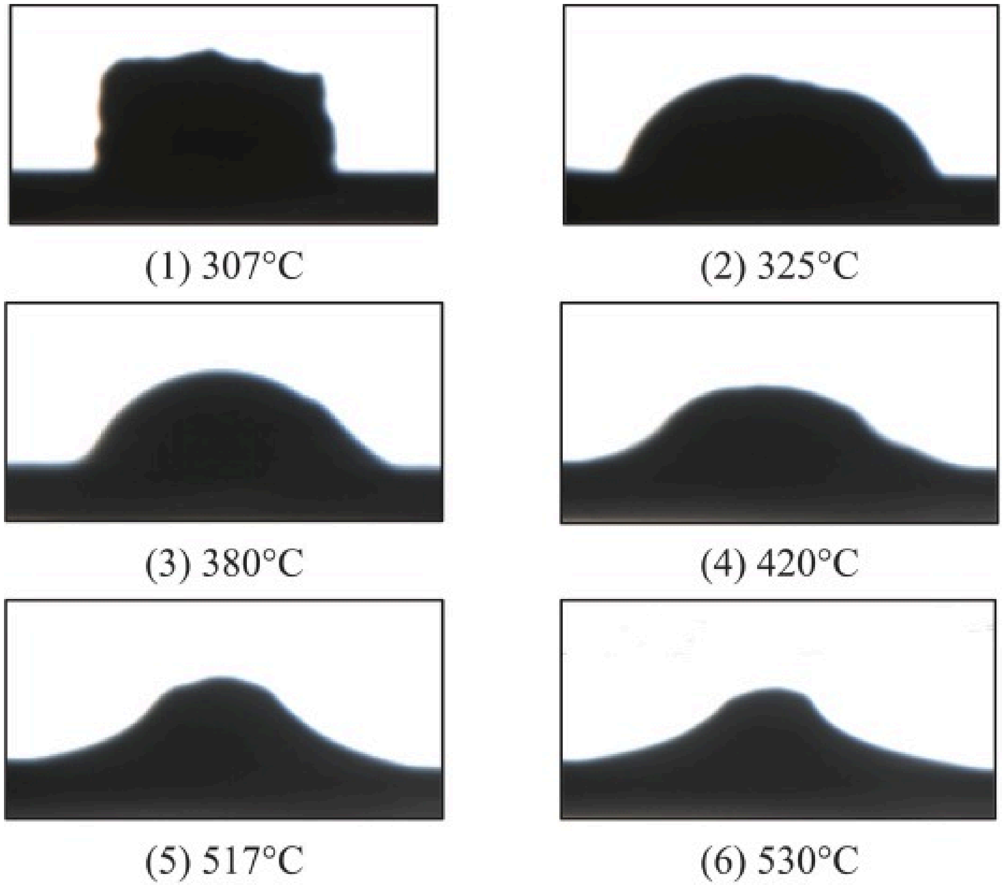

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Fig. 4. Images of sodium morphology on the stainless steel surface.

As shown in the Fig. 5, the contact angles of sodium obtained from present experiments are compared with those results from Bader [23] and Hodkin [20]. The contact angles are obtained with the tangent fitting method, which is pre calibrated with standard contact angle image. To ensure reproducibility, five measurements are taken for each sample, revealing the maximum deviation of less than 5° for angles above 90° and less than 3° for angles below 90°. The experimental results demonstrate a clear temperature-dependent wettability transition for liquid sodium on stainless steel surfaces. Below 300°C, the system remains in a non-wetting regime characterized by contact angles exceeding 90°. A distinct wetting transition occurs near 320°C, marking the onset of progressive interfacial adhesion. Above this critical temperature, the contact angle exhibits a consistent monotonic decrease with increasing temperature, ultimately reaching complete wetting conditions (<3°) at temperatures exceeding 530°C. This systematic evolution from non-wetting to complete wetting behavior reveals three characteristic regimes: (1) a temperature-insensitive non-wetting phase below 300°C, (2) a transitional wetting phase between 320–530°C, and (3) a complete wetting phase above 530°C, with the contact angle reduction rate being most pronounced in the transitional regime.

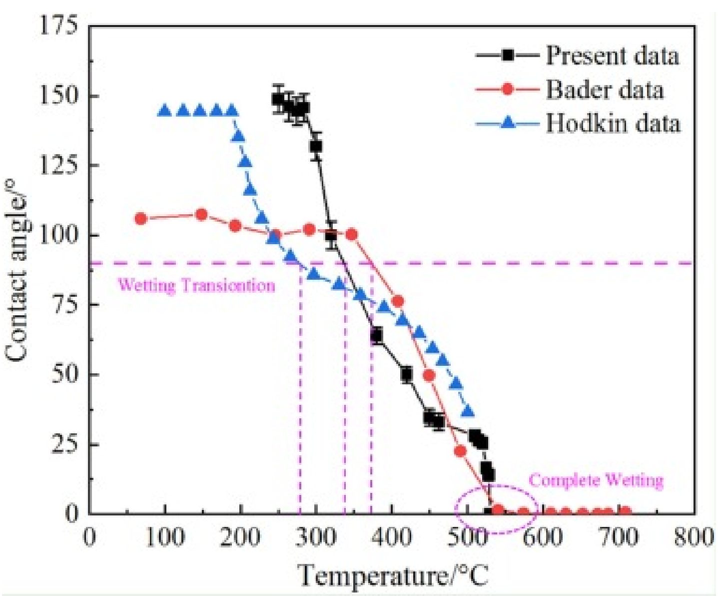

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Fig. 5. Effect of temperature on contact angle of liquid sodium on stainless steel.

Addison [7] pointed out that the Cr element in stainless steel will affect the wetting contact angle of liquid sodium on the surface of stainless steel, resulting in the higher temperature required for wetting than pure metal surface. While the dense oxide layer on stainless steel provides excellent corrosion resistance, it simultaneously inhibits wetting between the molten sodium and metal surface. However, the liquid sodium will undergo reduction reactions with the oxide layer at high temperatures. This redox process gradually exposes the underlying pure iron substrate, enabling the establishment of strong metallic bonding and consequent wetting. The reduction kinetics are temperature-dependent, with complete oxide removal resulting in optimal wetting conditions. The reaction equations are as follows:

$$Cr_2O_3 + 3Na \rightarrow 3NaCrO_2 + Cr \tag{1}$$

$$\mathrm{Cr}O_2 + Na \rightarrow NaCrO_2 \tag{2}$$

$$\mathrm{Cr}O_3 + 3Na \rightarrow NaCrO_2 + Na_2O \tag{3}$$

3.2. Experimental results with different oxide thicknesses

In order to quantitatively investigate the influence of oxide film thickness on wettability, the magnetron sputtering method is employed to deposit precisely controlled $\mathrm{Cr_2O_3}$ coatings (10μm, 20μm, 30μm, and 50μm) on stainless steel substrates. The experimental process was similar to the previous method and the captured wetting image are shown in the Fig. 6, which clearly demonstrates the progressive changes in interfacial phenomena across the different oxide thicknesses.

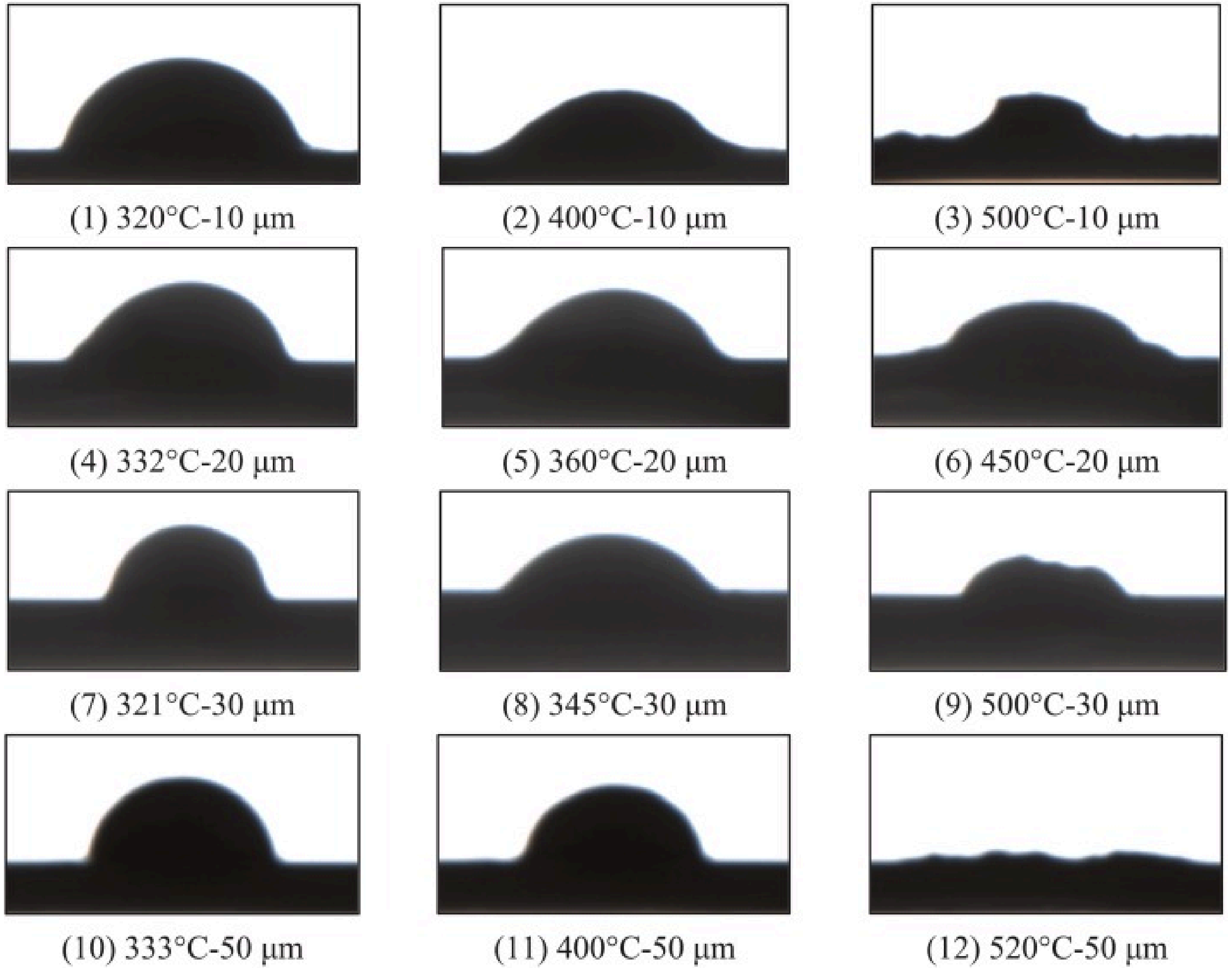

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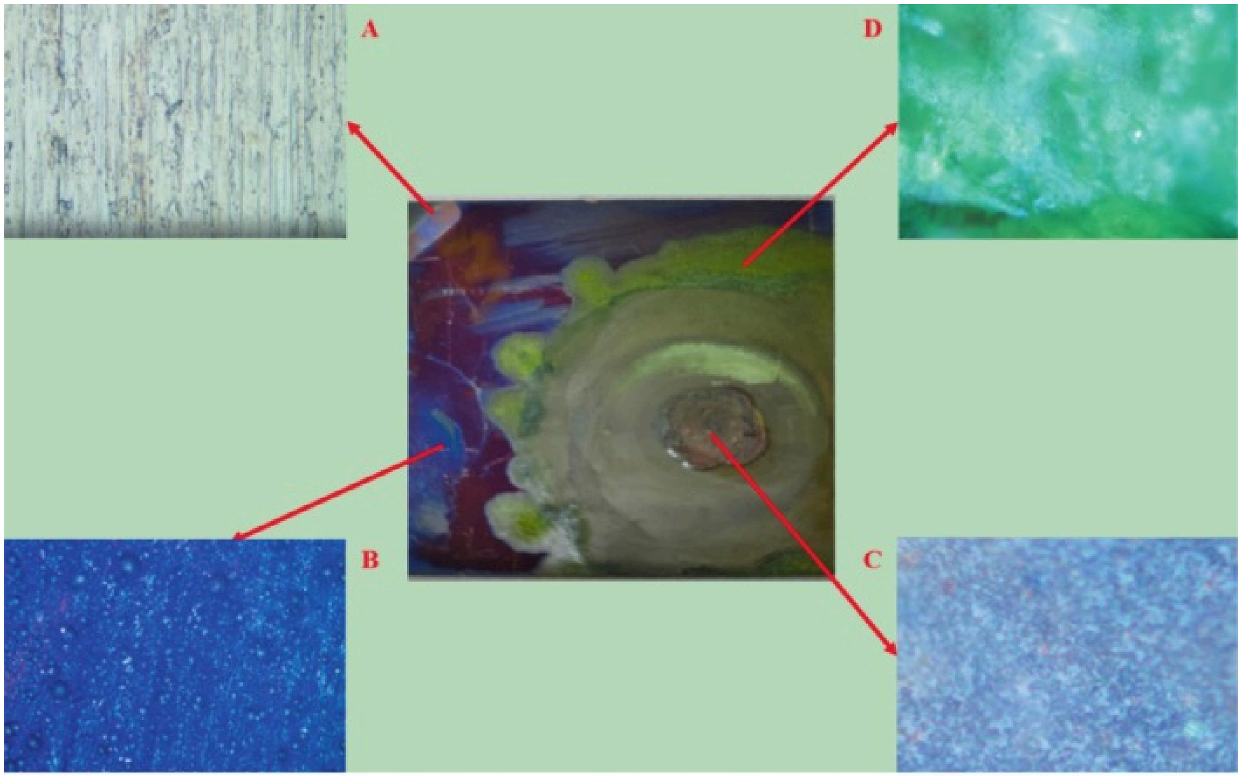
Fig. 6. Images of sodium morphology on stainless steel surface with different oxide thicknesses.

Fig. 7 presents the post-experimental surface morphology of the 30μm Cr_2O_3 coating stainless steel following liquid sodium exposure, with corresponding microstructural characterization at selected regions. It can be observed that a yellow brown circular area appeared in the plate central region, which is speculated to be the sodium oxide and the corrosion product generated by the reaction between liquid sodium and stainless steel. Surrounding this core region, a light green annular zone is observed, corresponding to the ternary oxide sodium chromite $NaCrO_2$ generated by the reaction between sodium and surface oxide. This microstructural evolution demonstrates the progressive corrosion mechanism, where initial sodium penetration through the oxide layer leads to complex reaction product generation.



A: stainless steel without coating; B: Cr_2O_3 coating;

C: stainless steel after reaction; D: $NaCrO_2$ produced by reaction

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Fig. 7. Different parts of the stainless steel surface with 30μm Cr_2O_3 coating after test.

The interfacial reaction between liquid sodium and Cr_2O_3-coated stainless steel exhibits a progressive corrosion pattern characterized by distinct zonal differentiation. Initial contact preferentially occurs at the central region where extended reaction duration promotes extensive formation of sodium-iron-oxide ternary compounds. This reacted zone shows significant morphological deviation from the original stainless steel surface due to comprehensive chemical transformation. The subsequent development of a light green annular region indicates that the unoxidized liquid sodium flows into the adjacent area and continues to react with the surrounding surface Cr_2O_3, marking the secondary reaction front. The production of $NaCrO_2$ is observed in the state of solid but soft crystalline structure and is not completely dissolved in the liquid sodium. Its existence in the liquid sodium system will also impose an effect of inhibiting the wettability. The observed spatial progression from central corrosion to peripheral reaction fronts demonstrates the complex interplay between reaction kinetics, wetting dynamics, and phase stability in the sodium-stainless steel system.

The variations of contact angles with increasing temperature for liquid sodium on stainless steel with different oxide thicknesses are compared in Fig. 8. It can be seen that the thickness of the oxide layer (10–50μm) has minimal effect on the final contact angle. The results demonstrate the distinct transition point between 320–335°C where the contact angle reaches 90°, marking the shift from non-wetting to wetting behavior. As temperature increases further, the contact angle progressively decreases to approximately 30° at 460°C. Complete wetting is ultimately achieved in the temperature range of 510–530°C, where liquid sodium fully spreads across the stainless steel surface. These findings indicate that while oxide thickness primarily affects reaction kinetics, the fundamental wetting transition remains predominantly temperature-dependent.

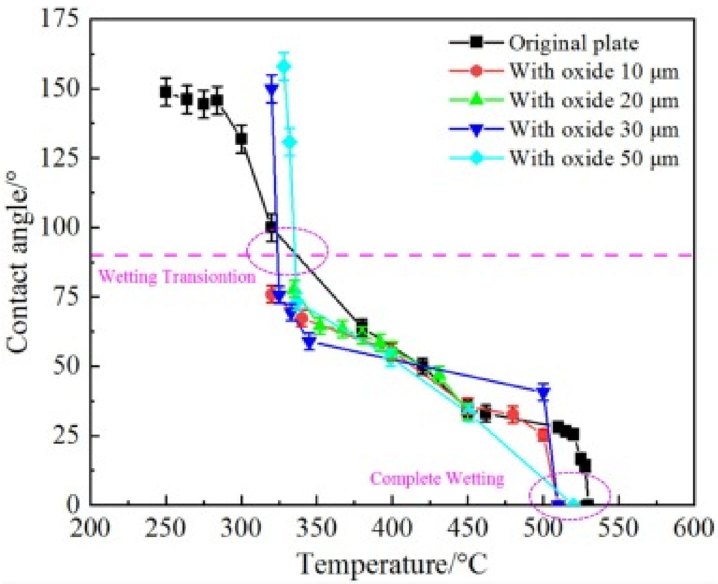

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Fig. 8. Effect of oxide thicknesses on the contact angle of liquid sodium on stainless steel.

4. Computational methodology

4.1. Model development

The molecular dynamics (MD) method is used for simulation of the interfacial mechanism of liquid sodium wetting behavior. As shown in Fig. 9, three types of surfaces are constructed: smooth stainless steel substrate, Cr_2O_3-coated surface, and rough surface with rectangular channels. These types are designed to align closely with the experimental specimens. The smooth stainless steel surface represents the unoxidized substrate in the experiments, allowing investigation of intrinsic wetting behavior under ideal conditions. The Cr_2O_3-coated surface corresponds to the magnetron sputtering sample, revealing the atomistic mechanism of wetting inhibition by the oxide layer. The rough surface with regular rectangular grooves serves as the simplified representation of machined or corroded surfaces, reflecting the Wenzel wetting regime. By integrating the simulation and experimental analyses, this study enables the multiscale understanding from macroscopic phenomena to microscopic mechanisms.

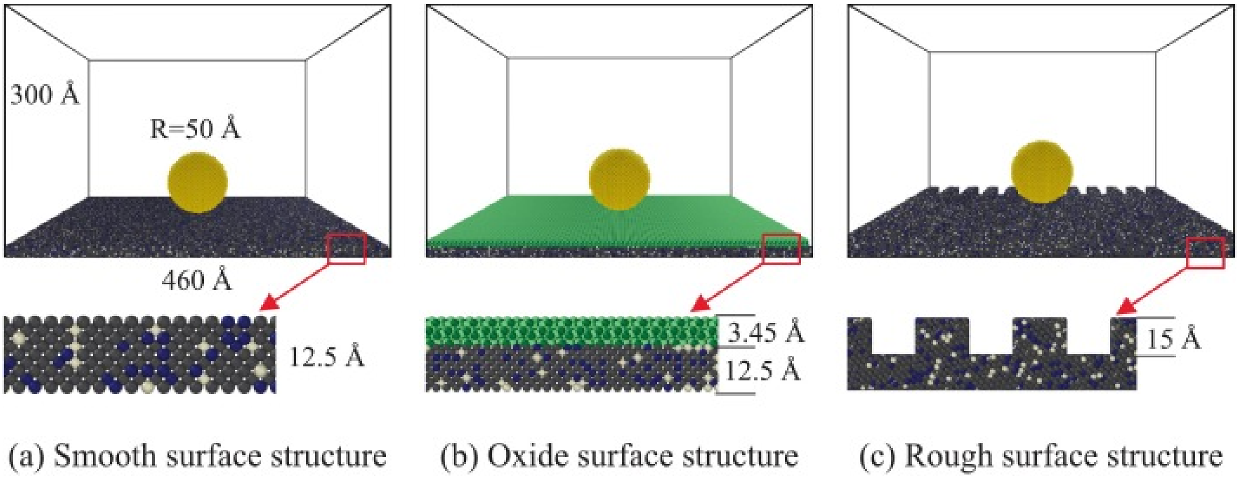

Fig. 9. Schematic diagram of three wetting structure models.

The smooth stainless steel substrate is constructed by replacing the atomic species with single crystals of iron in a proportional manner. The alloy composition is precisely formulated with three constituent elements (Fe, Cr, Ni) in a strictly controlled atomic ratio of 7:2:1 respectively. The size of the substrate is set to 460Å×460Å×12.5Å ($1Å=10^{-10}$ m), in which the initial configuration of the sodium droplet is a standard sphere with a radius of 50Å. The droplet is placed at 3Å directly above the substrate, and the length and width dimensions of the substrate are much larger than the radius of sodium droplet of 50Å. This design ensures that the droplet maintains a sufficient distance from the boundary during wetting and spreading to avoid the disturbance of atoms in the periodic boundary effectively.

For the Cr_2O_3-coated stainless steel system, the oxide coating is precisely constructed with a thickness of 3.45Å. The Cr_2O_3 layer is initially modeled using intrinsic parameters within Materials Studio software, followed by integration with the stainless steel substrate in LAMMPS to create a composite structure featuring an atomically well-defined interface. As for the textured surface with precisely controlled roughness parameters, periodic rectangular grooves are engineered onto the pristine stainless steel substrate, featuring the uniform depth of 15Å and the calculated average roughness (Ra) of 1.67Å. Within this simulation framework, the substrate and Cr_2O_3 coating atoms remain fixed in their lattice positions to maintain structural integrity, while permitting full dynamic evolution of the liquid sodium atoms. The complete atomic ensemble is contained within a vacuum-bound simulation cell with dimensions of 460Å×460Å×300Å.

The MD simulations are performed with time step of 1fs and cutoff radius of 10Å. The wetting process is simulated for 1ns (10^6 steps), ensuring adequate time for the liquid sodium to reach equilibrium spreading configurations. Before the simulation, the system is first relaxed under the NPT ensemble (constant temperature and pressure). After relaxation is completed, the wetting simulation is carried out using the NVT ensemble (constant temperature and volume). Temperature control is achieved using the Nosé–Hoover thermal bath method, with both isothermal and stepwise heating schemes. The isothermal scheme maintains the system at constant temperatures of 326.85°C (600K) and 626.85°C (900K) for 1ns. These two temperatures correspond to the critical range in the experiments where the transition of wetting and complete wetting occurs. This scheme is primarily used to investigate the influence of the Cr_2O_3 layer on the stainless steel surface on the contact angle and the formation of precursor film of liquid sodium. The stepwise heating scheme with variable temperature increments is shown in Fig. 10. It starts at 126.85°C (400K) and increases in 100°C increments every 40ps until reaching 526.85°C (800K), followed by 50°C increments every 40ps up to 626.85°C (900K). This covers the experimental heating range of 300–600°C and is used to study the dynamic spreading kinetics of liquid sodium under non-equilibrium conditions.

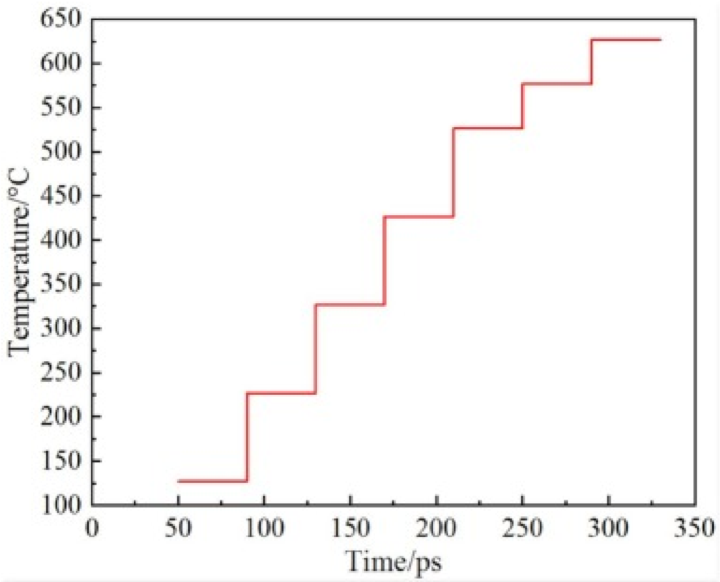

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Fig. 10. Stepped heating program during simulation.

4.2. Solution methods and potential functions

The MD method is fundamentally based on Newtonian classical mechanics, where the trajectories of all atoms are determined by numerically solving their equations of motion. The computational algorithm follows these key steps: (1) system initialization with predefined potential parameters and atomic positions, (2) force calculation for all particles using specified interatomic potentials, (3) numerical integration of Newton's equations, and (4) statistical averaging of thermodynamic quantities after system equilibration. For multi-particle systems, the coupled nonlinear differential equations governing atomic motions cannot be solved analytically, necessitating iterative numerical integration solutions. The typical algorithms used in MD simulation are Verlet algorithm, velocity Verlet algorithm, prediction correction algorithm, leapfrog algorithm and so on [29]. The Verlet algorithm, which is widely used, is selected for this simulation [30]. Eqs.

(4), (5), (6) represent the primary formulas of the Verlet algorithm, which are used to calculate the particle position, velocity, and acceleration, respectively:

$$r_i(t+\Delta t) = 2r_i(t) - r_i(t-\Delta t) + a_i(t)\Delta t^2 \tag{4}$$

$$v_i(t) = \frac{r_i(t+\Delta t) - r_i(t-\Delta t)}{2\Delta t} \tag{5}$$

$$a_i(t) = F_i(t) \,/\, m \tag{6}$$

The selection of appropriate interatomic potentials is paramount in MD simulations, as it fundamentally determines both the accuracy of results and computational efficiency. In this simulation of sodium droplet, the Embedded Atom Method (EAM), proposed by Daw and Baskes [31], is well-suited for describing atomic interactions in metallic systems. The total energy of the system can be expressed as:

$$E_{\mathrm{tot}} = \sum_{i=1}^{N} F_i \rho_i + \frac{1}{2}\sum_{i \neq j}^{N} \phi_{\mathrm{ij}} r_{\mathrm{ij}} \tag{7}$$

where ϕ_{ij} is the short-range pair potential function, F_i is the embedding energy, r_{ij} is the distance between two atoms, and ρ_i is the background electron density of the atoms at the location.

Nichol et al. [32] derived an EAM potential based on experimental data, which provides better description of sodium atoms and shows good agreement with physical properties of metallic sodium, such as density and melting point. Therefore, the sodium droplet in this simulation is described using this empirical potential function. For the interaction potential between sodium atoms and other atoms, the Lennard-Jones (L-J) potential function is adopted. The functional form of this model is given as follows:

$$u_{\mathrm{L\text{-}J}} = 4\varepsilon\frac{\sigma^{12}}{r} - \frac{\sigma^{6}}{r} \tag{8}$$

where r represents the distance between two atoms, while ε and σ are potential parameters that vary depending on the types of atoms involved. Specifically, ε denotes the depth of the potential well, corresponding to the energy difference between the minimum potential energy and zero, which characterizes the strength of the interatomic interactions. The parameter σ indicates the distance at which the interatomic potential equals zero, representing the equilibrium separation between atoms. For different types of atomic pairs, the potential parameters can be calculated using the Lorentz–Berthelot mixing rules, where the arithmetic mean is used to compute σ_{ij}, and the geometric mean is used to compute ε_{ij}. The specific equations are as follows:

$$\sigma_{\mathrm{ij}} = \frac{1}{2}\sigma_{\mathrm{ii}} + \sigma_{\mathrm{jj}} \tag{9}$$

$$\varepsilon_{\mathrm{ij}} = \sqrt{\varepsilon_{\mathrm{ii}}\varepsilon_{\mathrm{jj}}} \tag{10}$$

For the parameters of the L-J potential functions of sodium, iron, chromium, nickel, and oxygen elements involved in the model, many scholars have derived and fitted them by means of experimental comparisons and other approaches [[33], [34], [35], [36]]. Based on the literature, the ε and σ parameters for all elements in the simulation system are obtained. Combined with the above equations, the L-J potential parameters for atomic pairs in the system are listed in Table 1.

Table 1. Parameter of L-J potential function.

Element combination	Parameter values	
	σ (Å)	**ε (eV)**
Na-Fe	2.701	0.10682
Na-Cr	2.710	0.10432
Na-Ni	2.653	0.11970
Na-O	3.298	0.01485

5. Simulation results

5.1. Isothermal wetting

The isothermal wetting process of liquid sodium on stainless steel is investigated via MD simulations at 326.85 °C (600 K) and 626.85 °C (900 K). In these strongly interacting systems where high-energy solid surfaces contact reactive liquid metals, the simulations reveal the formation of a distinct precursor film at the solid–liquid interface. This thin film, extending ahead of the nominal contact line and approximately 1–2 molecular layers thick, indicates favorable wetting behavior in the system [37]. The presence and dynamics of this precursor film provide fundamental insights into the early stages of wetting and spreading in liquid metal-solid systems.

Fig. 11 presents the MD simulation results of the wetting process of liquid sodium on stainless steel surface without oxide layer as well as with Cr_2O_3 coating under isothermal conditions at 326.85 °C (600 K). With progressive simulation time, the sodium droplet exhibits gradual surface spreading accompanied by continuous contact angle reduction. On uncoated stainless steel surfaces, atomic-scale analysis identifies the emergence of a well-defined precursor film at the advancing contact line at 1.0 ns. In marked contrast, Cr_2O_3-coated surfaces demonstrate complete suppression of precursor film formation due to weakened interfacial bonding. This is consistent with the higher contact angles observed experimentally on the coated surface (Fig. 8). The droplet on the Cr_2O_3-coated surface maintains in curved cap shape with significantly delayed spreading rate and limited reduction in contact angle. This is because the weak interfacial interaction suppresses the precursor film, hindering the spreading process.

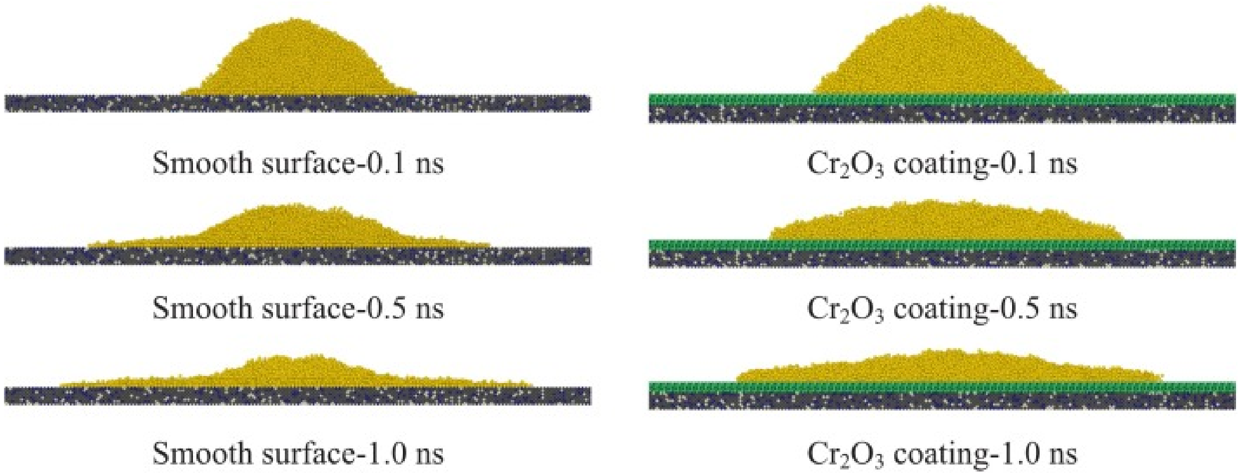

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Fig. 11. 326.85 °C (600 K) isothermal wetting process on the surface with and without oxide layer.

Fig. 12 shows the MD simulation results of the wetting process of liquid sodium on stainless steel surface without oxide layer as well as with Cr_2O_3 coating under isothermal conditions at 626.85 °C (900 K). It can be seen that the elevated temperature of 626.85 °C (900 K) dramatically enhances wetting kinetics relative to the 326.85 °C (600 K) isothermal condition, regardless of surface coating status. At 1.0 ns, liquid sodium on the smooth surface is almost fully spread. However, due to the initial inhibition on the coated surface, the spreading rate remains relatively delayed even though chemical reactions has began to occur. Further combining the top-view images of final spreading states shown in Fig. 13, it can be clearly observed that the spreading area on the smooth plate at 626.85 °C (900 K) is significantly larger than that with 326.85 °C (600 K). Additionally, the spreading is relatively uniform and regular, approximating a circular shape. At high temperatures, the spreading area of liquid sodium on the Cr_2O_3-coated surface also increases, but with lesser extent. Moreover, the liquid sodium exhibits irregular localized aggregation, forming an elliptical thickened cluster.

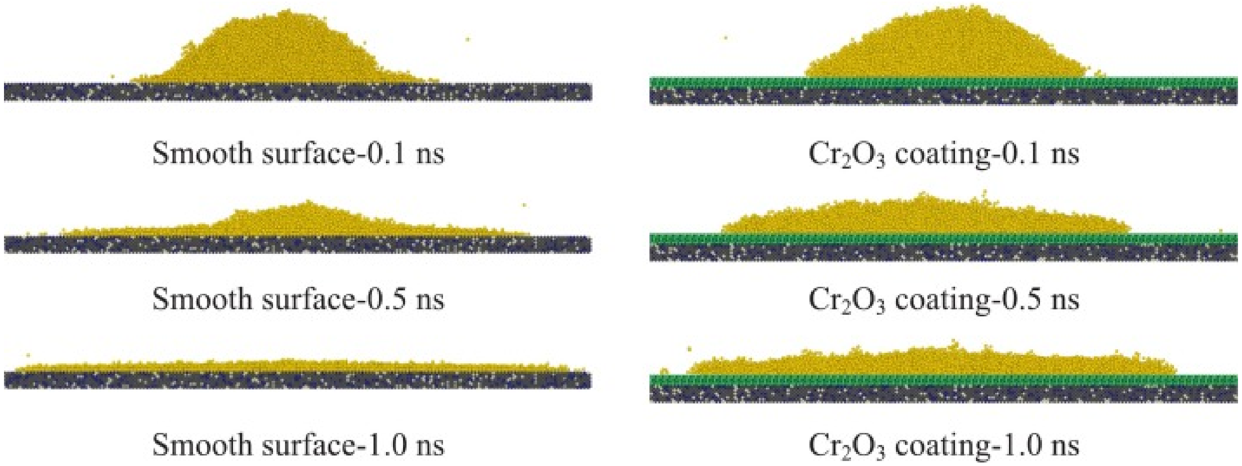

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Fig. 12. 626.85 °C (900 K) isothermal wetting process on the surface with and without oxide layer.

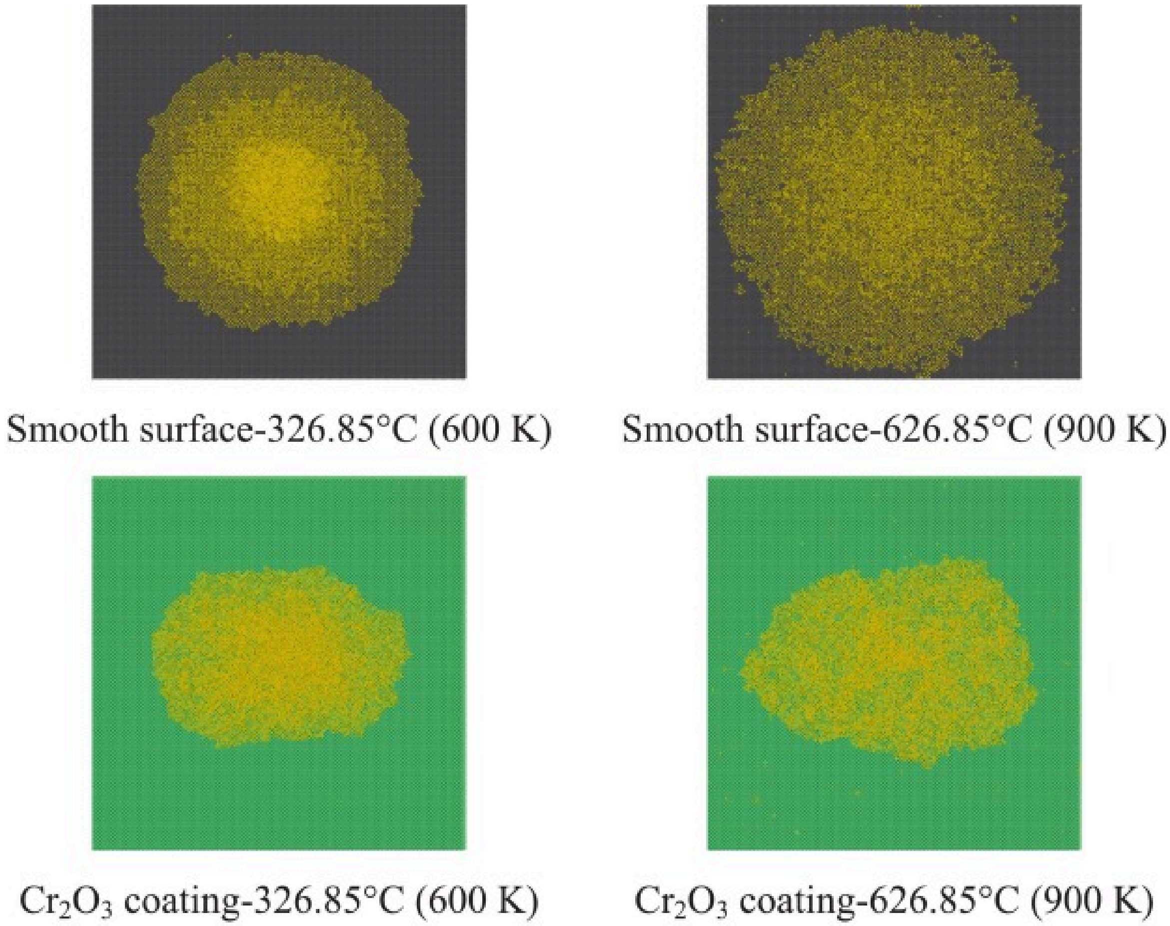

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Fig. 13. Top-view images of final spreading states on the surface with and without oxide layer.

Overall, the precursor film on the stainless steel surface is influenced by the wetting duration, wetting temperature, and the presence of oxide layer. Under isothermal wetting at 326.85 °C (600K), an obvious precursor film forms on the smooth surface compared to the chromium oxide surface. As compared in Fig. 14, Fig. 14, the wetting radius of liquid sodium at 326.85 °C (600K) on the smooth surface is significantly larger than that on the chromium oxide surface, with a maximum difference of 10.44%. At 626.85 °C (900K) isothermal wetting for 1 ns, liquid sodium has fully spread on the smooth surface, while no distinct precursor film is observed on the chromium oxide surface. Under the same conditions, the maximum difference in wetting radius between the smooth surface and the Cr_2O_3-coated surface increases to 15.56%. From Fig. 14, Fig. 14, it can be observed that compared with the Cr_2O_3-coated surface, the wetting temperature imposed more significant effect on the wetting radius of liquid sodium on the smooth surface, while its influence on the Cr_2O_3-coated surface is relatively limited.

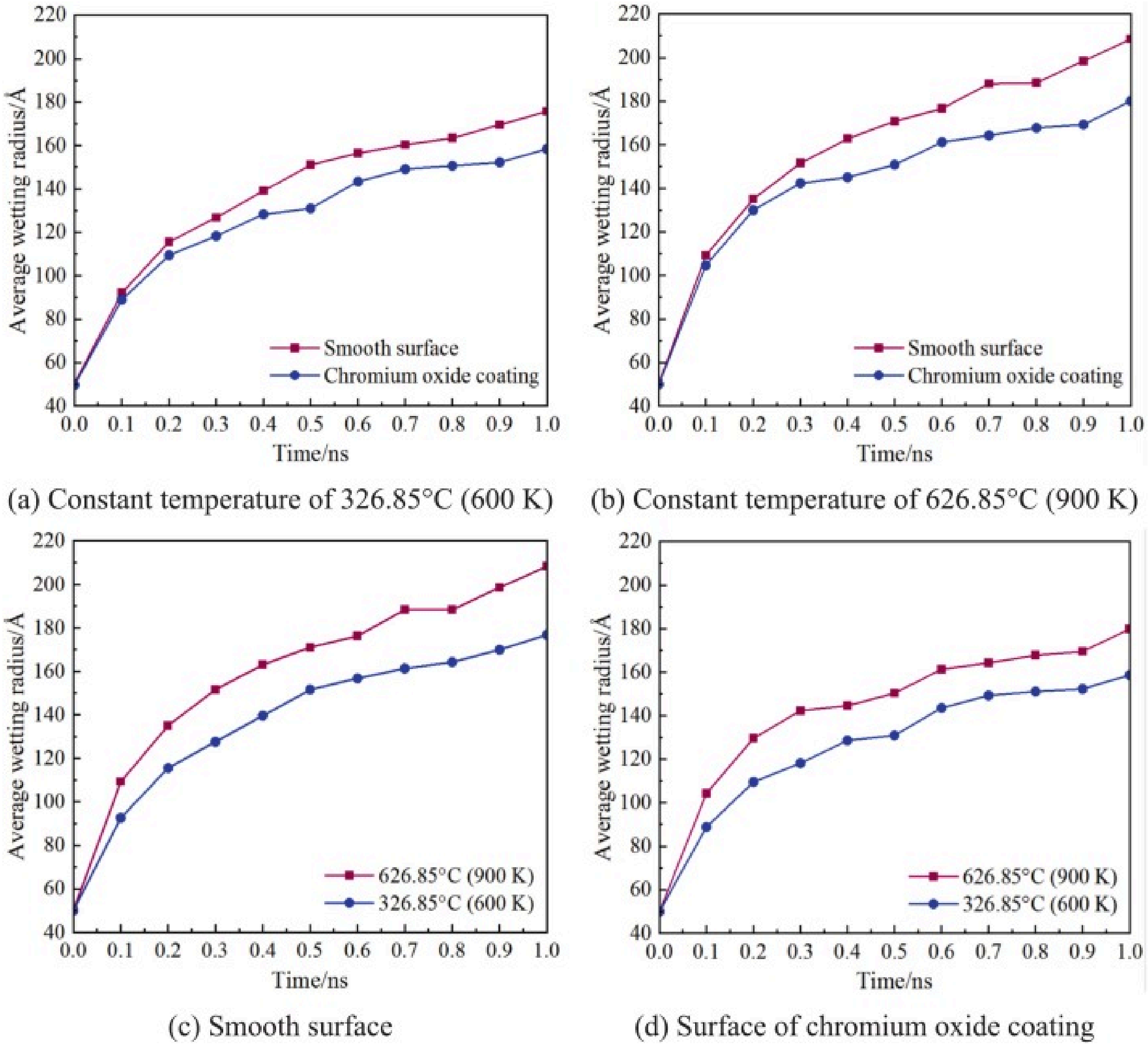

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Fig. 14. Variation of average wetting radius with the wetting duration.

5.2. Stepwise heating wetting

To further investigate the spreading and wetting behavior of liquid sodium on different surfaces, an additional simulation is conducted using the rough stainless steel surface with rectangular grooves and stepwise heating scheme. During spreading on the rectangular grooved surface, the sodium droplet exhibited the Wenzel's model. The spreading configurations at several temperature points are shown in Fig. 15. Fig. 16 shows the spreading state of liquid sodium on three different surfaces at 526.85°C (800K) temperature in front, side and top views. It can be clearly observed that the droplet spreads more uniformly and over a larger area on the smooth surface.

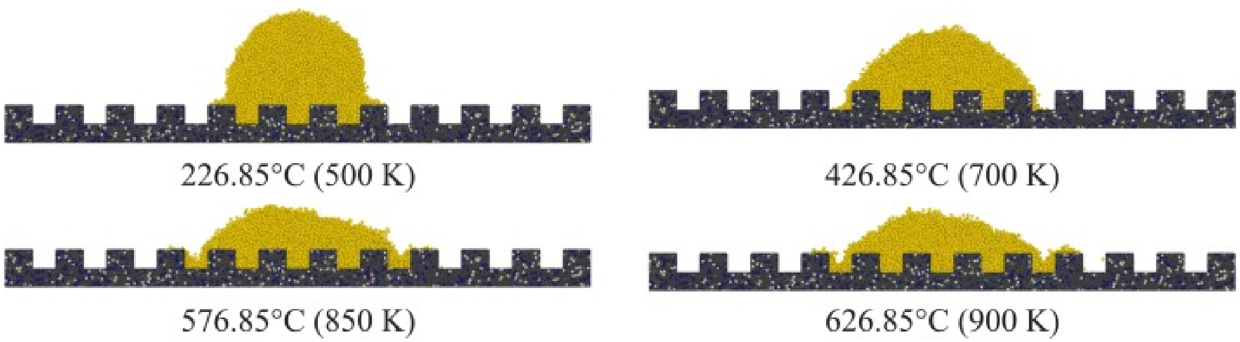

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Fig. 15. Spreading process on a rough surface with stepwise heating.

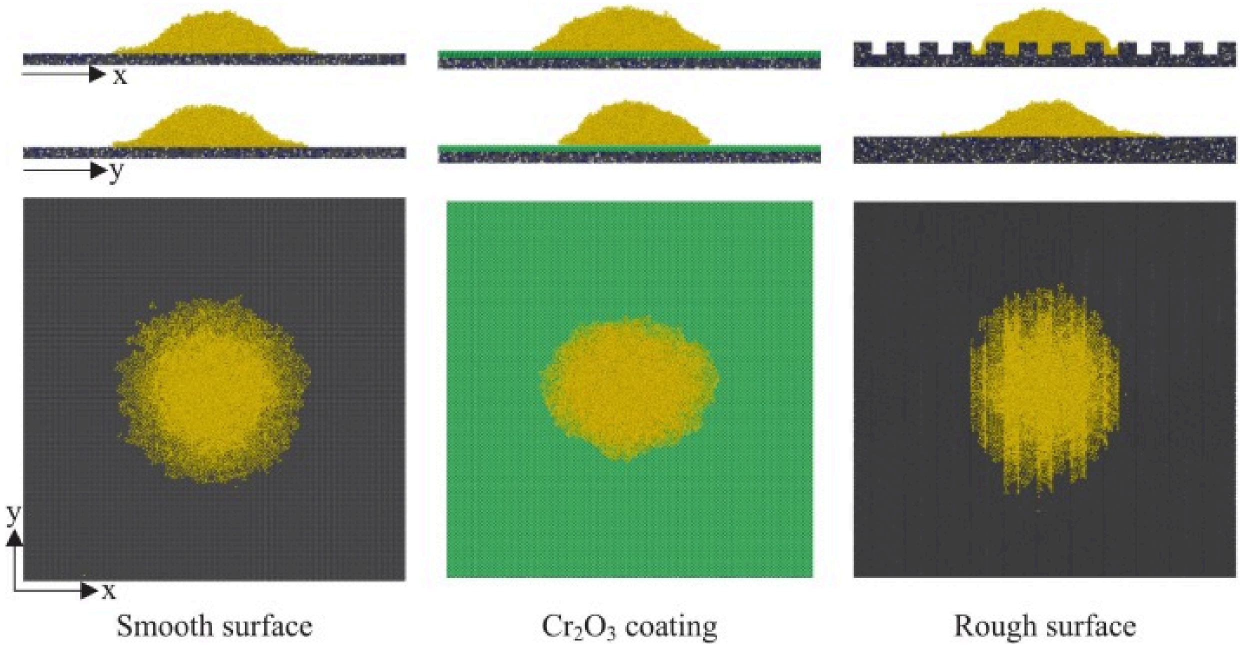

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Fig. 16. Spreading state of liquid sodium on different surfaces at 526.85 °C (800K).

Fig. 17 shows the variation in the average wetting radius at each temperature point. At each temperature point, the smooth surface exhibits the largest wetting radius, followed by the Cr_2O_3-coated surface. The rough surface presents the smallest wetting radius, although with relatively small difference compared to the Cr_2O_3-coated surfaces. The radius differences gradually widen with increasing temperatures.

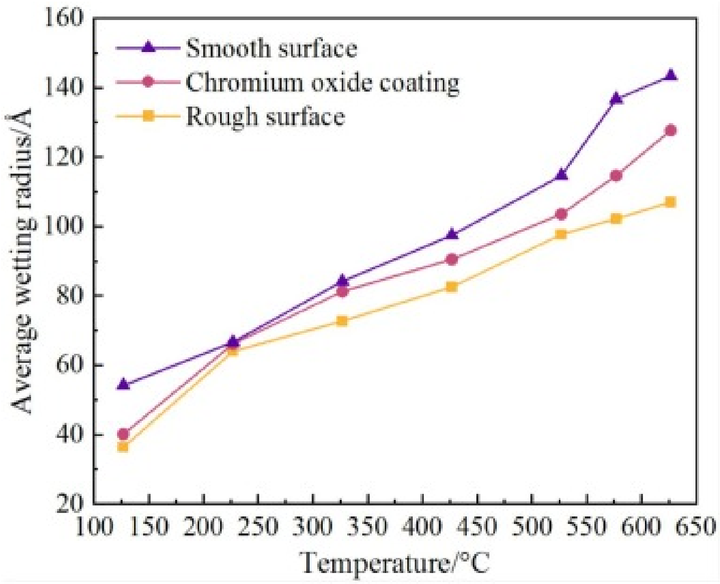

Fig. 17. Variation of average wetting radius with increasing temperature.

Fig. 18 compares the variation trends of the liquid sodium contact angle under the three conditions. It can be seen that the simulation successfully captures the monotonic decrease in contact angle with increasing temperature, which is in full agreement with the experimental measurements (Figs. 5 and 8). In addition, the contact angle on the Cr_2O_3-coated surface is consistently greater than that on the smooth surface. The contact angle on the rough surface becomes lower than that on the Cr_2O_3-coated surface after approximately 376.85 °C (650 K) and drops below that of the smooth surface after 576.85 °C (850 K). Therefore, in the temperature range lower than 326.85 °C (600 K), the wetting evolution mechanisms of the three surfaces tend to converge. At higher temperatures, strong intermolecular interactions on the smooth surface promote spreading, resulting in the largest wetting radius. On the Cr_2O_3-coated surface, both the contact angle and average spreading radius remain consistently lower than those on the smooth surface. In contrast, the rough surface constrained by Wenzel's model exhibits hindered actual spreading despite with decreased contact angle.

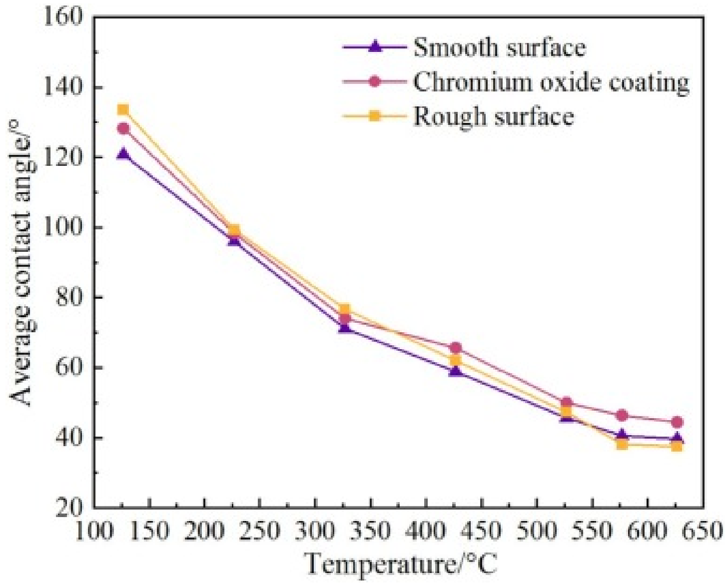

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Fig. 18. Variation of average contact angle with increasing temperature.

6. Conclusions

This study combines experiments and molecular dynamics simulations to investigate liquid sodium wetting on different kinds of stainless steel surfaces. Both macroscopic wetting behavior and atomic-scale interfacial kinetics are examined. The main conclusions are as follows:

(1) The interaction between liquid sodium and stainless steel surfaces is primarily determined by temperature-dependent chemical reactions with the oxide layer. At temperatures below 300°C, the stable oxide layer prevents any wetting behavior. As temperatures rise above this threshold,

the oxide layer undergoes progressive reduction, allowing sodium to penetrate the surface and eventually achieve complete wetting.

(2) The wetting process exhibits clear temperature-dependent stages. The initial wetting transition begins around 320–335 °C, marked by the contact angle reaching 90°. Further heating to 460 °C reduces the contact angle to approximately 30°, with complete wetting occurring in range of 510–530 °C.

(3) The development of precursor films, crucial for wetting initiation, is significantly affected by surface conditions. Cr_2O_3 coating effectively suppresses precursor film formation, resulting in smaller wetting radii compared to bare surfaces. This inhibition becomes more pronounced at higher temperatures, with wetting radius differences increasing from 10.44% at 326.85 °C to 15.56% at 626.85 °C.

(4) Cr_2O_3 coating consistently maintains higher contact angles than uncoated surface under all tested conditions. On textured surface, sodium exhibits characteristic Wenzel wetting behavior, showing limited lateral spreading but enhanced contact angle reduction at elevated temperatures.

CRediT authorship contribution statement

Can Wang: Writing – original draft, Software, Investigation, Formal analysis. **Yugao Ma:** Supervision, Methodology, Conceptualization. **Langlang Tian:** Validation, Investigation, Software. **Luteng Zhang:** Writing – original draft, Funding acquisition, Conceptualization, Investigation, Methodology, Writing – review & editing. **Simiao Tang:** Investigation, Funding acquisition, Formal analysis. **Zaiyong Ma:** Methodology, Data curation. **Qiang Lian:** Software, Methodology. **Wan Sun:** Validation, Software, Methodology. **Longxiang Zhu:** Validation, Formal analysis. **Liangming Pan:** Supervision, Project administration, Methodology. **Meiyue Yan:** Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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